Arpit Suryavanshi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.44
GURU NANAK PUBLIC SCHOOL	Class XII (CBSE)	2024	80.20%
GURU NANAK PUBLIC SCHOOL	Class X (CBSE)	2022	93.00%

SCHOLASTIC ACHIEVEMENTS

- Selected among **35 students nationwide** for the prestigious **National Initiative on Undergraduate Science (NIUS) Astronomy Program, HBCSE-TIFR**, following a competitive national selection process.

INTERNSHIP

Aryabhatta Research Institute of Observational Sciences *(May 2026 - July 2026)*

- Engineered a 5-stage preprocessing framework converting 24,000+ spectra into 1201-dimensional ML-ready features.
- Trained XGBoost and 1D CNN across 8 classes, handling 115× imbalance using weighted learning.
- Achieved 88.5% test accuracy with XGBoost, outperforming the CNN by 10.8 percentage points, with a reproducible framework designed to scale to 54K+ spectra and incorporate latent-parameter estimation.

SkillEdu Varsity *(May 2025)*

- Mentored 50 students in designing and providing technical guidance through hands-on STEM activities.
- Delivered an interactive lecture on Supernovae and conducted live telescope observation sessions, simplifying advanced astronomy concepts for school students.

PROJECTS

Bayesian Gravitational Wave Parameter Estimation

- Built a **Bayesian framework** using Metropolis-Hastings MCMC to estimate **3 signal parameters** from noisy time-series data.
- Executed **400K MCMC iterations**, retaining **300K posterior samples** after burn-in for parameter estimation.
- Validated **convergence and parameter uncertainty** using FFT-based autocorrelation and **Effective Sample Size (ESS)** diagnostics, alongside posterior and waveform reconstruction analysis.

Option Pricing & Trading Edge Analysis

- Developed an **options pricing framework** using Black-Scholes, **Geometric Brownian Motion**, and Monte Carlo.
- Simulated **100,000+ trades** across **2 market regimes** over **252-day** price paths, evaluating profitability and P&L.
- Compared **win rates, variance, percentile distributions, and cumulative returns** across market regimes to quantify trading edge, assess buyer vs. seller performance, and identify profitability patterns under differing market conditions.

X-Ray Timing Analysis & QPO Detection

- Developed a **4-stage time-series framework** for X-ray light curves using **FFT-based Power Density Spectra (PDS)** analysis.
- Implemented **power-law and Lorentzian fitting** with **3 statistical validation techniques** for periodic signal detection, residual analysis, and uncertainty estimation.

SKILLS

Programming Languages: Python, C, C++, MySQL

Machine Learning: Scikit-learn, CNN, XGBoost, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Feature Engineering, Data Preprocessing, Model Evaluation, SciPy, Clustering, Regression, Optimization

Statistical & Computational Methods: Bayesian Inference, Markov Chain Monte Carlo (MCMC), Monte Carlo Simulation, Time-Series Analysis, Fourier Analysis, Statistical Modeling, Model Validation, Numerical Methods

Scientific Computing & Research Tools: Jupyter Notebook, Google Colab, VS Code, LaTeX, Git, GitHub, RDBMS

POSITIONS OF RESPONSIBILITY

Secretary of Academic Affairs, Student Gymkhana, IIT Hyderabad | *April 2026 - Present*

- Led a 100-member team across 17+ departments, representing 5,500+ students and managing 1,500+ voter elections.
- Drove institute-wide policy reforms with the Director, Dean & HODs, securing approval for a Statistics Minor, 10% AI/ML curriculum & 6th-semester internships.

Head, Cepheid - Astronomy Club, IIT Hyderabad | *May 2025 - April 2026*

- Led a 40-member team, driving technical, outreach, and faculty-mentored research initiatives across the club.
- Engaged 500+ participants through National Space Day, stargazing and technical events, while hosting researchers from Harvard, TIFR, SMU & University of Utah.

Student Representative -- Department Undergraduate Committee (DUGC) | *May 2025 - April 2026*

- Advocated AI/ML curriculum integration and resolved student concerns, coordinating with the HOD, faculty advisors, CRs, and students on academic improvements.

Student Coordinator -- Office of Career Services (OCS), IIT Hyderabad | *July 2025 - April 2026*

- Coordinated campus recruitment operations for 10+ companies, facilitating communication between recruiters, institute stakeholders, and 100+ students to ensure seamless execution of placement and internship processes.

EXTRACURRICULAR

- Conducted **outreach workshops** and telescope observation sessions for **multiple schools**, promoting STEM education.
- Won **5 Gold Medals** across **Cosplay (2×), Filmmaking, Stage Play, and Improvisation** at **Milan** and **Diesta**, IIT Hyderabad.
- Volunteered in **campus cleanliness drives**, contributing to institute-wide sustainability initiatives.